MATH 2102 Linear Algebra II

Chapter 2
Linear Transformations

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Chapter 2: Linear Transformations

A1. Setting the scene

assumptions: ①~③

Throughout this chapter, ① U, V, W denote vector spaces, where ② the field of scalars is $\mathbb{F}$ unless the content suggests otherwise.

Recall also that we have assumed ③ (all vector spaces to be finite-dimensional,) unless otherwise specified or when an explicit infinite-dimensional example is given. Note however that many of the definitions and results will apply to infinite-dimensional vector spaces, except those statements and proofs that involve bases and dimensions.

A2. Definition

We are familiar with linear transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$. More generally, we can define such functions between vector spaces.

A function $T : V \to W$ is said to be a linear transformation (or linear map) if

- $T(\mathbf{u} + \mathbf{v}) =$ ___T(u) + T(v)___ for all $\mathbf{u}, \mathbf{v} \in V$;
- $T(c\mathbf{v}) =$ ___c · T(v)___ for all $c \in \mathbb{F}$ and $\mathbf{v} \in V$.

We denote by $\mathcal{L}(V, W)$ the set of linear transformations from V to W.

A3. Examples

Some examples of linear transformations between vector spaces (other than $\mathbb{F}^n$) are shown below:

- $R : P_3(\mathbb{R}) \to P_2(\mathbb{R})$ defined by $R(p(x)) = p'(x)$

- $S : \mathcal{C}(\mathbb{R}) \to \mathbb{R}$ defined by $S(f(x)) = \int_0^1 f(x)\, dx$

- $T : \mathbb{F}_2^\infty \to \mathbb{F}_2^\infty$ defined by $T((x_1, x_2, x_3, \ldots)) = (x_2, x_3, x_4, \ldots)$

A4. Extension by linearity

We are familiar with the following result for linear transformations $T : \mathbb{R}^n \to \mathbb{R}^m$. The same is true for linear transformations between general vector spaces:

> A linear transformation $T : V \to W$ is uniquely determined by its values on a basis of V.
>
> More precisely, if $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for V and $\mathbf{w}_1$, $\mathbf{w}_2$, ..., $\mathbf{w}_n$ are (not necessarily distinct) vectors in W, there exists a unique $T \in \mathcal{L}(V, W)$ for which $T(\mathbf{v}_i) = \mathbf{w}_i$ for all i.
>
> In particular, we must have $T(\mathbf{0}) = \mathbf{0}$.

A5. Kernel and image

Let $T \in \mathcal{L}(V, W)$. We are particularly interested in one subset of V and one subset of W that are actually subspaces.

	Subspace of V	Subspace of W
Name	kernel (or null space)	image (or range)
Notation	ker T (or null T)	im T (or range T)
Meaning	Pre-image of $\mathbf{0}_W$	Range of T
Mathematical definition	$\{v \in V : T(v) = 0\}$	$\{T(v) : v \in V\}$
Dimension	nullity	rank
Notation for dimension	nullity T = dim ker/null T	rank T

A5. Kernel and image

Recall the following linear transformations. Can you find their kernels and images?

- $R : P_3(\mathbb{R}) \to P_2(\mathbb{R})$ defined by $R(p(x)) = p'(x)$

  $\ker R = P_0(\mathbb{R}) = \mathbb{R}$

  $\operatorname{im} R = P_2(\mathbb{R})$

- $S : \mathcal{C}(\mathbb{R}) \to \mathbb{R}$ defined by $S(f(x)) = \int_0^1 f(x)\,dx$

  $\ker S = \{f(x) : \int_0^1 f(x)\,dx = 0\}$

  $\operatorname{im} S = \mathbb{R}$

- $T : \mathbb{F}_2^\infty \to \mathbb{F}_2^\infty$ defined by $T((x_1, x_2, x_3, \dots)) = (x_2, x_3, x_4, \dots)$

  $\ker T = \{(0,0,\dots), (1,0,0,\dots)\}$

  $\operatorname{im} T = \mathbb{F}_2^\infty$

A6. The fundamental theorem

We have the following fundamental theorem:

Rank-nullity theorem

For any $T \in \mathcal{L}(V, W)$, we have

$$\dim V = \operatorname{rank} T + \operatorname{nullity} T.$$

Proof:

Let $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ be a basis of $\ker T$. This can be extended to a basis of V, say, by adding the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, ..., $\mathbf{v}_n$.

let $\{u_1 \ldots u_m\}$ is a basis of $\ker T$

and $\{u_1 \ldots u_m\} \cup \{v_1 \ldots v_n\}$ is the basis V,

we want to show that $\{T(v_1) \ldots T(v_n)\}$ is basis of W.

we assume that: are linear dependent

i.e. $\exists$ $\sum c_i T(v_i) = 0$

$\therefore$ $T(\sum c_i v_i) = 0$

i.e. $\sum c_i v_i$ in $\ker T$

A7. Injectivity and surjectivity

$\forall a, b,\ f(a) = f(b) \Rightarrow a = b$

Let $T \in \mathcal{L}(V, W)$.

- T is injective if and only if $\ker T = \underline{\ \{0\}\ }$; $\dim V = \operatorname{rank} T \leq \dim W$
- f^{-1} exist
- T is surjective if and only if $\operatorname{im} T = \underline{\ W\ }$. $\dim V \geq \dim W = \operatorname{rank} T$

Given that $\dim V = \operatorname{rank} T + \operatorname{nullity} T$, how would $\dim V$ and $\dim W$ compare in each case?

A7. Injectivity and surjectivity

Let $T \in \mathcal{L}(V, W)$, where $\dim V = \dim W$. Then the following statements are equivalent.

$\rightarrow \text{nullity}\, T = 0$

$\dim V = \text{rank}\, T = \dim W$

- T is injective. $\rightarrow$
- T is surjective. $\rightarrow \dim W = \text{rank}\, T = \dim V$
- T is bijective.

Chapter 2: Linear Transformations

B1. The space $\mathcal{L}(V, W)$

Recall that the set of all linear transformations $T : V \to W$ is denoted by $\mathcal{L}(V, W)$.

We often abbreviate $\mathcal{L}(V, V)$ as $\mathcal{L}(V)$, and elements in $\mathcal{L}(V)$ are called linear operators (or simply operators) on V. Several of the upcoming chapters will focus on studying operators on V.

$\mathcal{L}(V, W)$ is a vector space over $\mathbb{F}$:

- Addition is defined by $(S+T)\vec{v} = Sv + Tv$ for all $S, T \in \mathcal{L}(V, W)$;
- Scalar multiplication is defined by $(\lambda T) v = \lambda\, T(v)$ for all $\lambda \in \mathbb{F}$ and $T \in \mathcal{L}(V, W)$;
- The zero element of $\mathcal{L}(V, W)$ is $0(v) = 0, \forall v \in V$.

B1. The space $\mathcal{L}(V, W)$

$$\dim(\mathcal{L}(V, W)) = \dim V \ \underline{\ \times\ }\ \dim W$$

Proof:

Let $\{\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_n\}$ be a basis for V and $\{\mathbf{w}_1, \mathbf{w}_2, ..., \mathbf{w}_m\}$ be a basis for W. We construct a basis for $\mathcal{L}(V, W)$ as follows:

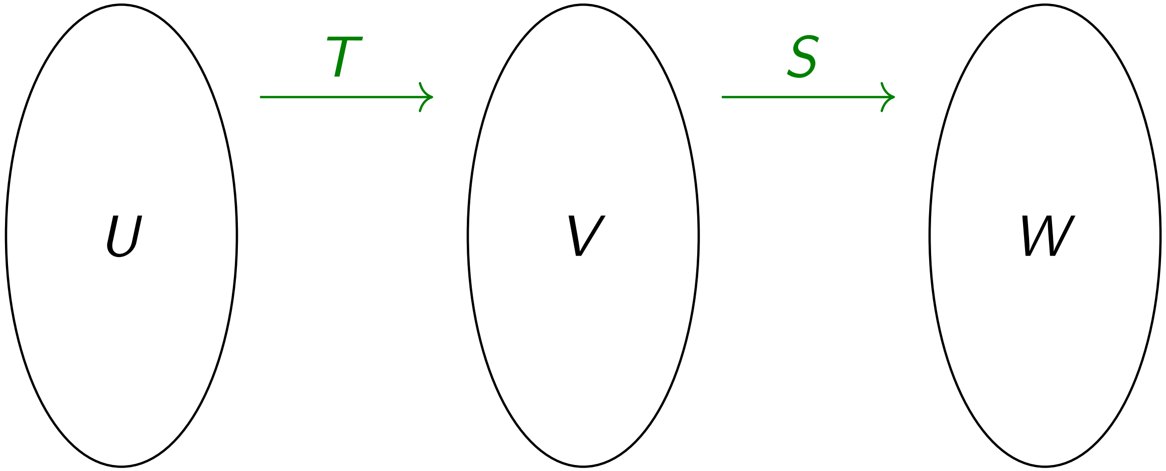
$$T_{ij}(v_k) = \begin{cases} w_j & , \text{if } i = k \\ 0 & , \text{otherwise} \end{cases}$$

B2. Composition of linear transformations

Apart from addition and scalar multiplication, we can also perform composition on linear transformations. More importantly, we have the following result.

Let $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$. Then $S \circ T \in \mathcal{L}(U, W)$.

Note that $S \circ T$ is sometimes abbreviated as ST.



B3. Inverse of linear transformations

Apart from composition, another operation that creates new linear transformations from existing ones will be inversion. A bijective linear transformation is invertible.

Suppose $T \in \mathcal{L}(V, W)$ is invertible. Then $T^{-1} \in \mathcal{L}(W, V)$.

B3. Inverse of linear transformations

Prove the following. Does it remind you of what you learnt about matrices?

Let $T \in \mathcal{L}(V, W)$ and $S \in \mathcal{L}(W, V)$, where $\dim V = \dim W$. Then $ST = \mathrm{id}_V$ if and only if $TS = \mathrm{id}_W$.

Proof:

Suppose $ST = \mathrm{id}_V$. Then T is injective since

we assume that $\ker T \neq \{0\}$. let $v \neq 0 \in \ker T$

$\therefore Tv = 0$, $T0 = 0$

$v = STv = S0 = 0$ ✗

B4. Isomorphism of vector spaces

Compare $\mathbb{R}^3$ and $P_2(\mathbb{R})$. In what sense are they 'the same'?

$P_2(R) = \vec{w}^T (1, x, x^2)$

$w \in R^3$

B4. Isomorphism of vector spaces

Formally, we make the following definition.

V and W are said to be isomorphic vector spaces, denoted $V \cong W$, if there exists an invertible linear transformation $T : V \to W$ (which is called an isomorphism, or more precisely, 'vector space isomorphism').

i.e. one to one

bijective

i.e. $\mathbb{R}^3 \cong P_2(\mathbb{R})$

$(1,2,5) \to x^2+2x+5$

$(0,1,1) \to x+1$

B4. Isomorphism of vector spaces

We have the following simple characterisation of finite-dimensional vector spaces.

If $\dim V = n$, then $V \cong \mathbb{F}^n$.

Let $\{\vec{v_i}\}$ is basis of V.

then $\forall \vec{v} \in V$, $\exists \sum c_i \vec{v_i} = \vec{v}$

$(c_1, \cdots c_n) \in \mathbb{R}^n \rightarrow \sum c_i \vec{v_i}$

B5. Quotient space revisited

Suppose W is a subspace of V. Recall that the quotient space V/W is defined by

$$V/W = \{\mathbf{v} + W : \mathbf{v} \in V\}.$$

This naturally induces the projection map $\pi : V \to V/W$ defined by

$$\pi(\mathbf{v}) = \mathbf{v} + W.$$

It is easy to check that this is a surjective linear map.

In view of this, we have

$$\dim(V/W) = \dim V \;\underline{\quad - \quad}\; \dim W.$$

B6. Isomorphism theorem

Let $T \in \mathcal{L}(V, W)$. Define a function $\widetilde{T} : V/(\ker T) \to W$ by

$$\widetilde{T}(\mathbf{v} + \ker T) = T(\mathbf{v}).$$

Then $\widetilde{T}$ is a linear transformation.

What's special about $\widetilde{T}$? Explore using the linear transformation $T : \mathbb{F}^3 \to \mathbb{F}^2$ defined by

$$T(x, y, z) = (x - y, x - z).$$

B6. Isomorphism theorem

First Isomorphism Theorem for Vector Spaces

Let $T \in \mathcal{L}(V, W)$. Then

$$V / \ker T \cong \operatorname{im} T.$$

Chapter 2: Linear Transformations

C1. Review

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. We know that there exists a unique $m \times n$ matrix A, called the standard matrix of T, such that

$$T(\mathbf{x}) = A\mathbf{x} \qquad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

More generally, for any ordered basis $\mathcal{B}$ of $\mathbb{R}^n$ and $\mathcal{C}$ of $\mathbb{R}^m$, there exists a unique matrix, which can be denoted by $[T]_{\mathcal{B}\mathcal{C}}$ or $\mathcal{M}_{\mathcal{B}\mathcal{C}}(T)$, such that

$$[T(\mathbf{x})]_{\mathcal{C}} = [T]_{\mathcal{B}\mathcal{C}}[\mathbf{x}]_{\mathcal{B}} \qquad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

C2. Extension to general vector spaces

Let's see what happens in general vector spaces. Recall the example $R : P_3(\mathbb{R}) \to P_2(\mathbb{R})$ defined by

$$R(p(x)) = p'(x).$$

Take the bases $\mathcal{B} = \{1, x, x^2, x^3\}$ for $P_3(\mathbb{R})$ and $\mathcal{C} = \{1, x, x^2\}$ for $P_2(\mathbb{R})$. How can we represent R using a matrix?

$$[R]_{\{1,x,x^2,x^3\}\{1,x,x^2\}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

C2. Extension to general vector spaces

More generally, suppose $T \in \mathcal{L}(V, W)$. Take the bases $\mathcal{B}$ for V and $\mathcal{C}$ for W. There exists a unique matrix, which can be denoted by $[T]_{\mathcal{BC}}$ or $\mathcal{M}_{\mathcal{BC}}(T)$, such that

$$[T(\mathbf{v})]_{\mathcal{C}} = [T]_{\mathcal{BC}}[\mathbf{v}]_{\mathcal{B}}.$$

$B = \{b_1, \dots b_n\}$

$C = \{c_1, \dots c_n\}$

$[T]_{BC} = \begin{bmatrix} [c_1]_B & [c_2]_B & \cdots & [c_n]_B \end{bmatrix}$

Some remarks on the notations:

- We may drop the reference to $\mathcal{B}$ and $\mathcal{C}$ (e.g. write $\mathcal{M}(T)$) if there is no danger of confusion.
- When $V = W$ and $\mathcal{B} = \mathcal{C}$, we may also simplify $[T]_{\mathcal{BC}}$ as $[T]_{\mathcal{B}}$ and $\mathcal{M}_{\mathcal{BC}}(T)$ as $\mathcal{M}_{\mathcal{B}}(T)$.

C3. Operations on linear transformations vs. operations on matrices

Let $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$. Assume we have fixed bases for V and W, whose sizes are n and m respectively. Then $\mathcal{M}(S), \mathcal{M}(T) \in \mathbb{F}^{m,n}$.

It is easy to check that the following equalities hold:

- $\mathcal{M}(S + T) = \mathcal{M}(S) + \mathcal{M}(T)$
- $\mathcal{M}(\lambda T) = \lambda \mathcal{M}(T)$

Thus $\mathcal{M} : \mathcal{L}(V, W) \to \mathbb{F}^{m,n}$ is a vector space isomorphism.

C3. Operations on linear transformations vs. operations on matrices

Furthermore, we have the following results, assuming we have fixed bases for the corresponding vector spaces.

- If $T \in \mathcal{L}(V, W)$ and $S \in \mathcal{L}(U, V)$, then $\mathcal{M}(TS) = \mathcal{M}(T)\mathcal{M}(S)$.
- If $T \in \mathcal{L}(V)$ is invertible, then $\mathcal{M}(T^{-1}) = (\mathcal{M}(T))^{-1}$.

C4. Change of basis

Recall that for $R : P_3(\mathbb{R}) \to P_2(\mathbb{R})$ defined by $R(p(x)) = p'(x)$, by taking bases $\mathcal{B} = \{1, x, x^2, x^3\}$ and $\mathcal{C} = \{1, x, x^2\}$, we have

$$[T]_{\mathcal{B}\mathcal{C}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

If we take alternative bases $\mathcal{B}' = \{1,\ 2+x,\ x^2,\ x^2 - x^3\}$ and $\mathcal{C}' = \{1,\ 2x,\ 3x^2\}$, what is $[T]_{\mathcal{B}'\mathcal{C}'}$? What is the relationship between $[T]_{\mathcal{B}\mathcal{C}}$ and $[T]_{\mathcal{B}'\mathcal{C}'}$?

C4. Change of basis

I of V

Fix a basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ of V. Then clearly $[\mathrm{id}_V]_{\mathcal{B}} = I_n$.

How about $[\mathrm{id}_V]_{\mathcal{B}\mathcal{B}'}$, where $\mathcal{B}' = \{\mathbf{v}_1', \mathbf{v}_2', \ldots, \mathbf{v}_n'\}$ is another basis of V? We have

$[\mathrm{id}_V[v_1]]_{B'}$

$$[\mathrm{id}_V]_{\mathcal{B}\mathcal{B}'} = \left(\begin{array}{c|c|c|c} \uparrow & \uparrow & \uparrow & \uparrow \\ [\mathbf{v}_1]_{\mathcal{B}'} & [\mathbf{v}_2]_{\mathcal{B}'} & \ldots & [\mathbf{v}_n]_{\mathcal{B}'} \\ \downarrow & \downarrow & \downarrow & \downarrow \end{array}\right).$$

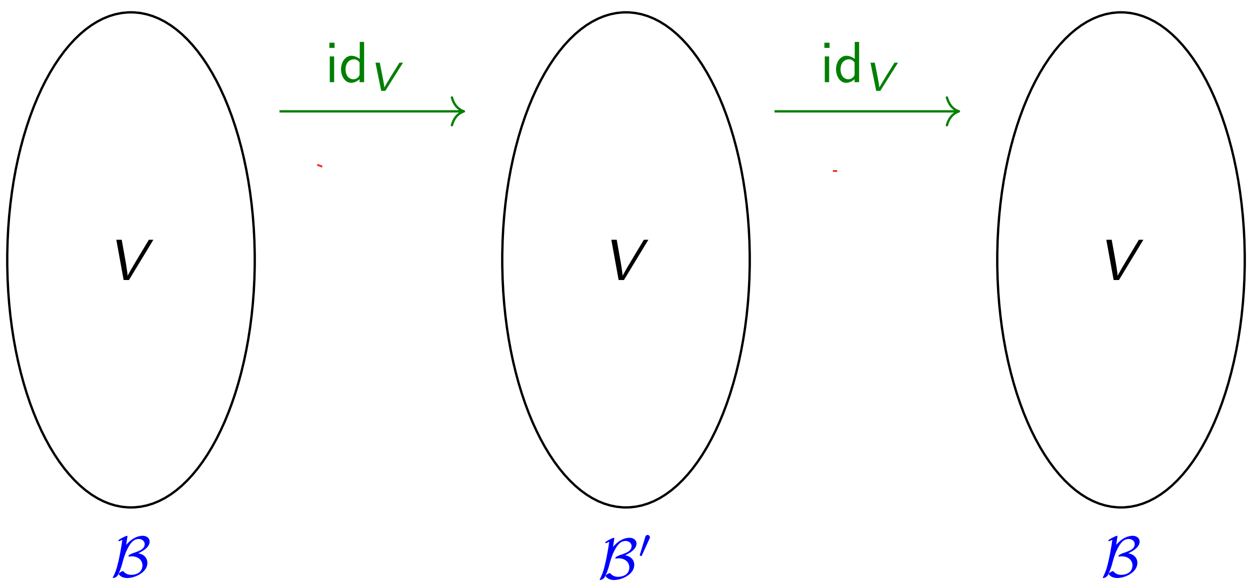
This is sometimes called the change of basis matrix from $\mathcal{B}$ to $\mathcal{B}'$. For instance, for $\mathcal{B} = \{1, x, x^2, x^3\}$ and $\mathcal{B}' = \{1,\ 2+x,\ x^2,\ x^2 - x^3\}$, we have

$$[\mathrm{id}_V]_{BB'} = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

C4. Change of basis

It easily follows that

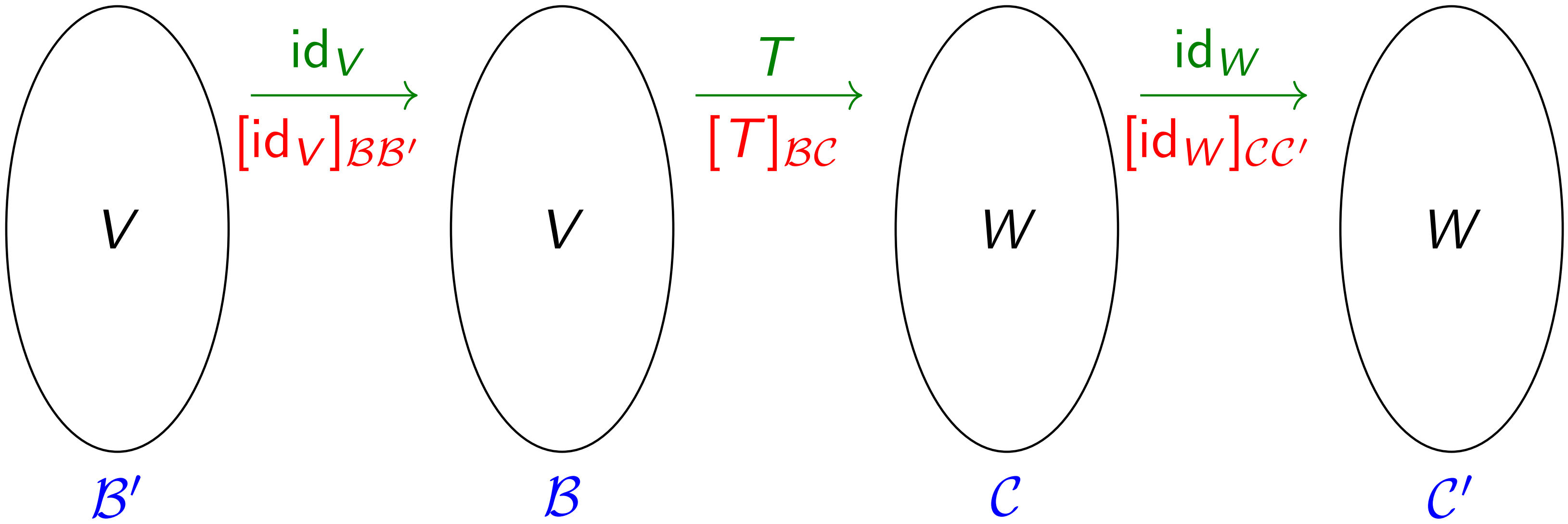
- $[\mathbf{v}]_{\mathcal{B}'} = [\mathrm{id}]_{\mathcal{B}\mathcal{B}'}[\mathbf{v}]_{\mathcal{B}}$
- $[\mathrm{id}]_{\mathcal{B}\mathcal{B}'}[\mathrm{id}]_{\mathcal{B}'\mathcal{B}} = I$



C4. Change of basis

We can now answer the previous question regarding $[T]_{\mathcal{B}\mathcal{C}}$ and $[T]_{\mathcal{B}'\mathcal{C}'}$. We have

$$[T]_{\mathcal{B}'\mathcal{C}'} = [\mathrm{id}_W]_{\mathcal{C}\mathcal{C}'}[T]_{\mathcal{B}\mathcal{C}}[\mathrm{id}_V]_{\mathcal{B}'\mathcal{B}}$$



What happens if $V = W$, $\mathcal{B} = \mathcal{C}$ and $\mathcal{B}' = \mathcal{C}'$?

Chapter 2: Linear Transformations

D1. Examples for motivation

Most linear transformations we have seen so far map vectors to vectors. How about functions that map vectors to scalars?

Consider the following examples:

- $f : \mathbb{F}^3 \to \mathbb{F}$ defined by $f(x, y, z) = 4x - 5y + 2z$
- $f : \mathbb{C}[x] \to \mathbb{C}$ defined by $f(p(x)) = 3p''(5) + 7p(4)$
- $f : \mathcal{C}(\mathbb{R}) \to \mathbb{R}$ defined by $f(g(x)) = \displaystyle\int_0^1 g(x)\, dx$

These are actually linear transformations from a vector space to $\mathbb{F}$.

D2. Dual space

We make the following definitions.

- A linear transformation from V to $\mathbb{F}$ is called a linear functional on V.
- The set of all linear functionals on V, i.e. $\mathcal{L}(V, \mathbb{F})$, is called the dual space (or simply dual) of V, denoted by V' (or in some books, V^*, but we shall not use this notation).

D2. Dual space

Recall that $\mathcal{L}(V, W)$ is a vector space over $\mathbb{F}$. Hence $V' = \mathcal{L}(V, \mathbb{F})$ is also a vector space over $\mathbb{F}$. We can thus ask a number of questions.

- How to perform addition and scalar multiplication on V'?
- What is the zero vector in V'?
- What is the dimension of V'?

D3. Dual basis

How to find a basis for $\mathcal{L}(V, W)$? How to find a basis for $V' = \mathcal{L}(V, \mathbb{F})$?

Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a basis for V. For each $i \in \{1, 2, ..., n\}$, define $\delta_i : V \to \mathbb{F}$ by

$$\delta_i(\mathbf{v}_j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}.$$

Then $\mathcal{B}' = \{\delta_1, \delta_2, \ldots, \delta_n\}$ is called the dual basis of $\mathcal{B}$.

D3. Dual basis

As naturally expected,

$\mathcal{B}' = \{\delta_1, \delta_2, \ldots, \delta_n\}$ forms a basis for V'.

Proof:

It suffices to show that δ_1, δ_2, ..., δ_n are linearly independent since $\underline{n = \dim V'}$.

Suppose $c_1\delta_1 + c_2\delta_2 + \cdots + c_n\delta_n = \mathbf{0}$. Then

let $f = \sum c_i \delta_i$

then $f v_1 = \cdots = f v_n = 0$

i.e. $c_1 = \cdots = c_n = 0$

D3. Dual basis

The following is one application of the dual basis. Suppose $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis for V. Then every $\mathbf{v} \in V$ can be expressed as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, ..., $\mathbf{v}_n$. Can you fill in the coefficients with the help of the dual basis?

$$\mathbf{v} = \underline{\delta_1(v)}\,\mathbf{v}_1 + \underline{\delta_2(v)}\,\mathbf{v}_2 + \cdots + \underline{\delta_n(v)}\,\mathbf{v}_n$$

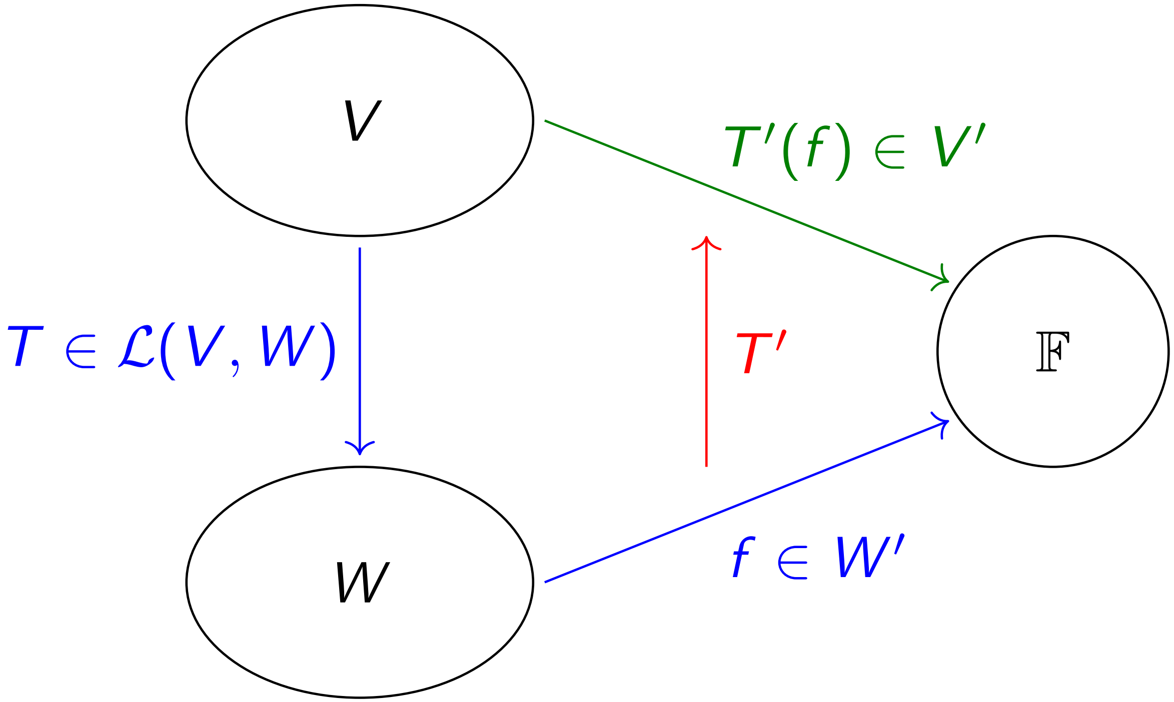
D4. Dual map

Let $T \in \mathcal{L}(V, W)$. Define $T' : W' \to V'$ by

$V \to$ $W \to F$

$$T'(f) = f \circ T.$$

Such T' is called the dual map of T.



D4. Dual map

As an example, let T be the differential operator on $\mathbb{R}[x]$, i.e.

$$T(p(x)) = \frac{d}{dx}[p(x)] \qquad \text{for all } p(x) \in \mathbb{R}[x].$$

Find $T'(f)$ in each of the following cases.

(a) $f(p(x)) = p(3)$

$T'(f) = f \circ T = p'(3)$

(b) $f(p(x)) = \int_0^1 p(x)\, dx$

$T'(f) = f \circ T$

$= \int_0^1 p'(x)\, dx$

$= p(1) - p(0)$

D4. Dual map

Let $T \in \mathcal{L}(V, W)$. Then

(1) $T' \in \mathcal{L}(W', V')$

D4. Dual map

$(T+S)'(f) = f\circ(T+S)$

$= f\circ T + f\circ S$

Let $T \in \mathcal{L}(V, W)$. Then

(2) $(T+S)' = T' + S'$ for any $S \in \mathcal{L}(V, W)$

(3) $(\lambda T)' = \lambda T'$ for any $\lambda \in \mathbb{F}$

These imply that the map from $\mathcal{L}(V, W)$ to $\mathcal{L}(W', V')$ taking T to T' is linear.

D4. Dual map

Let $T \in \mathcal{L}(V, W)$. Then

$V \to W \to U \to F$ $\quad V' \to W' \to U'$

(4) $(S \circ T)' = T' \circ S'$ for $S \in \mathcal{L}(W, U)$

$(S\circ T)'(f) = f\circ(S\circ T)$

$= f\circ S\circ T = f(S(T(v)))$

$(T'\circ S')(f) = T'(S'(f))$

$= f\circ S\circ T$

$5(2x+y) - 2(5x-4y)$

$= 4x-3y$

$12-15$

$$\begin{tikzcd}
V \arrow[r, "T"] \arrow[rr, bend left, "S\circ T"] & W \arrow[r, "S"] & U
\end{tikzcd}$$

$V' \xleftarrow{T'} W' \xleftarrow{S'} U'$

F

$T'\circ S'$

Chapter 2: Linear Transformations

E1. Matrix representation of the dual map

Let $T \in \mathcal{L}(V, W)$. Recall that the dual map $T' : W' \to V'$ is defined by

$$T'(f) = f \circ T.$$

Suppose we take bases $\mathcal{B}$ of V and $\mathcal{C}$ of W. A natural question is how $\mathcal{M}_{\mathcal{C}'\mathcal{B}'}(T')$ is related to $\mathcal{M}_{\mathcal{B}\mathcal{C}}(T)$.

Try with the differential operator $T : P_3(\mathbb{R}) \to P_2(\mathbb{R})$.

E1. Matrix representation of the dual map

Let $T \in \mathcal{L}(V, W)$, $\mathcal{B}$ be a basis for V and $\mathcal{C}$ be a basis for W. Then

$$\mathcal{M}_{\mathcal{C}'\mathcal{B}'}(T') = [\mathcal{M}_{\mathcal{B}\mathcal{C}}(T)]^T$$

Proof:

Denote the elements of $\mathcal{B}$, $\mathcal{B}'$, $\mathcal{C}$ and $\mathcal{C}'$ by $\mathbf{v}_i$, f_i, $\mathbf{w}_j$ and g_j respectively, where $1 \leq i \leq n$ and $1 \leq j \leq m$. For arbitrary i, j, we compute $(g_j \circ T)(\mathbf{v}_i)$ in two different ways:

E2. Annihilator

Let $U \subseteq V$. Define the annihilator of U to be

$$U^0 = \{f \in V' : f(\mathbf{u}) = 0 \text{ for all } \mathbf{u} \in U\}.$$

In words, U^0 consists all linear functionals f on V such that $\ker f$ contains U. It is easy to check that U^0 is always a subspace of V'.

E2. Annihilator

Find U^0 in the following examples.

(a) $V = \mathbb{R}^5$

$U = \text{Span}\{\mathbf{e}_1, \mathbf{e}_2\}$

$U^0 = \text{span}\{\delta_3, \delta_4, \delta_5\}$.

(b) $V = \mathbb{R}[x]$

$U = \{p \in \mathbb{R}[x] : p(0) = 0\}$

$U^0 = \{ f(p(x)) = c, c \in \mathbb{R} \}$.

$U_0 = \{$

E2. Annihilator

If U is a subspace of V, then we have the following.

- $\dim U^0 = \dim V$ $\underline{\ -\ }$ $\dim U$.
- $U^0 = \{\mathbf{0}\}$ if and only if $U =$ $\underline{\ V\ }$
- $U^0 = V'$ if and only if $U =$ $\underline{\ \{0\}\ }$

E3. Annihilator and dual map

Let $T \in \mathcal{L}(V, W)$. Then we have the following properties about T'.

(1) The kernel of T' is $(\operatorname{im} T)^0$ and hence has dimension $\dim W - \operatorname{rank} T$.

(2) T' is injective if and only if T is surjective.

$T'(f) = 0 \to \forall v \in V,\ f(T(v)) = 0$

i.e. $f \in (\operatorname{im} T)^0$

E3. Annihilator and dual map

Let $T \in \mathcal{L}(V, W)$. Then we have the following properties about T'.

$f(T'(v))$

(3) The image of T' is $\underline{(\ker T)^0}$ and hence has dimension ____________.

(4) T' is surjective if and only if T is $\underline{\text{injective}}$.

$g \in W'$, $g \in \operatorname{im} T' \to g = f(T(\vec{v}))$

E4. An application of duality

We could use the previous results to give an alternative proof of the following result from MATH2101.

> The row space and column space of a matrix have the same dimension.

Proof:

Let $A \in \mathbb{F}^{m,n}$ and $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ whose standard matrix is A. Then

$A = [Te_1 . Te_2 \cdots]$

$\left.\begin{array}{l} \text{col } A = \text{span}\{Te_i\} = \text{im}\, T \\ \text{row } A = \text{col } A^T = \text{im}\, T' \end{array}\right\}$ same dim.